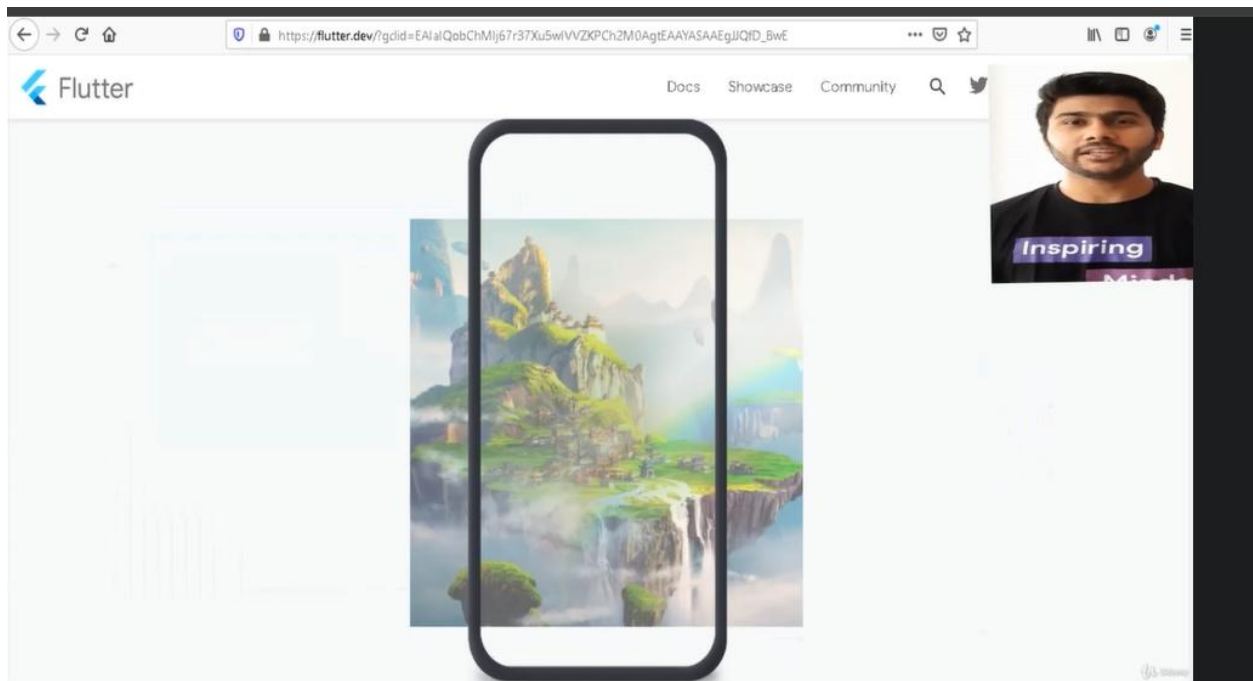
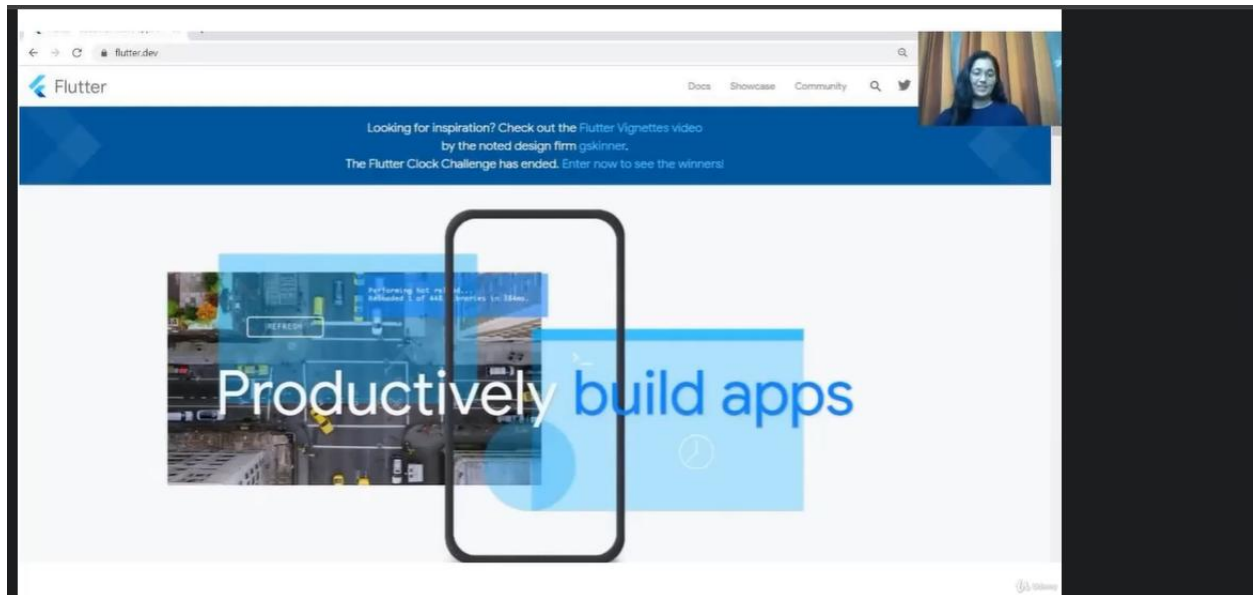


Tema 1. Introduccion



Tema 2. Instalación de Flutter (configuración del entorno)

The screenshot shows the Flutter documentation website. The top navigation bar includes 'Flutter', 'Documentos', 'Guías', 'Aprender', 'AI', and 'Referencia'. A search icon and a 'Comience' button are on the right. A purple banner at the top reads '¡Ayúdanos a mejorar Flutter! Participa en nuestra encuesta del segundo trimestre.' The left sidebar lists various topics under 'Comience' and 'Interfaz de usuario'. The main content area is titled 'Instalar Flutter manualmente' with a sub-header 'Aprende a instalar y configurar manualmente el SDK de Flutter.' Below this is a 'Consejo' (Tip) box suggesting VS Code for a simplified experience. The section 'Elige tu plataforma de desarrollo' (Choose your development platform) lists four options: Windows, macOS, Linux, and ChromeOS, each with its respective logo.

Flutter Documentos Guías Aprender AI Referencia

Comience

¡Ayúdanos a mejorar Flutter! Participa en nuestra encuesta del segundo trimestre.

instalar > A mano

Instalar Flutter manualmente

Aprende a instalar y configurar manualmente el SDK de Flutter.

Aprende a instalar y configurar manualmente tu entorno de desarrollo de Flutter.

Consejo

Si lo que buscas es instalar Flutter rápidamente, considera [instalarlo con VS Code](#) para una experiencia de configuración simplificada.

Elige tu plataforma de desarrollo

Las instrucciones de esta página están configuradas para explicar la instalación de Flutter en un dispositivo **Windows**.

Si desea seguir las instrucciones para un sistema operativo diferente, seleccione una de las siguientes opciones.

Windows macOS Linux ChromeOS

En esta página

- Elige tu plataforma de desarrollo
- Descargue el software necesario.
- Instalar y configurar Flutter
- Agrega Flutter a tu PATH
- Continúa tu viaje con Flutter

The screenshot shows the Flutter documentation website, specifically the 'Instalar y configurar Flutter' (Install and configure Flutter) page. The top navigation bar is the same as the previous screenshot. The left sidebar lists various topics under 'Más allá de la interfaz de usuario' and 'Recursos'. The main content area is titled 'Instalar y configurar Flutter' with a sub-header 'Para instalar el SDK de Flutter, descargue el paquete más reciente del archivo del SDK y, a continuación, extraiga el SDK en la ubicación donde desee guardarlo.' Below this is a numbered list of steps: 1. Descarga el paquete del SDK de Flutter (Download the Flutter SDK package), 2. Crea una carpeta para almacenar el SDK (Create a folder to store the SDK), and 3. Extraer el SDK (Extract the SDK). A 'Nota' (Note) box is also present, advising to choose a location without special characters or spaces. The bottom of the page shows the beginning of the first step, '1. Copia el siguiente comando.' (Copy the following command.)

Flutter Documentos Guías Aprender AI Referencia

Comience

Instalar y configurar Flutter

Para instalar el SDK de Flutter, descargue el paquete más reciente del archivo del SDK y, a continuación, extraiga el SDK en la ubicación donde desee guardarlo.

- 1 Descarga el paquete del SDK de Flutter**

Descarga el siguiente paquete de instalación para obtener la última versión estable del SDK de Flutter.

flutter_windows_3.44.2-stable.zip
- 2 Crea una carpeta para almacenar el SDK.**

Cree o busque una carpeta para almacenar el SDK extraído. Considere crear y usar un directorio en `%USERPROFILE%\develo` (`C:\Users\{username}\develo`).

Nota

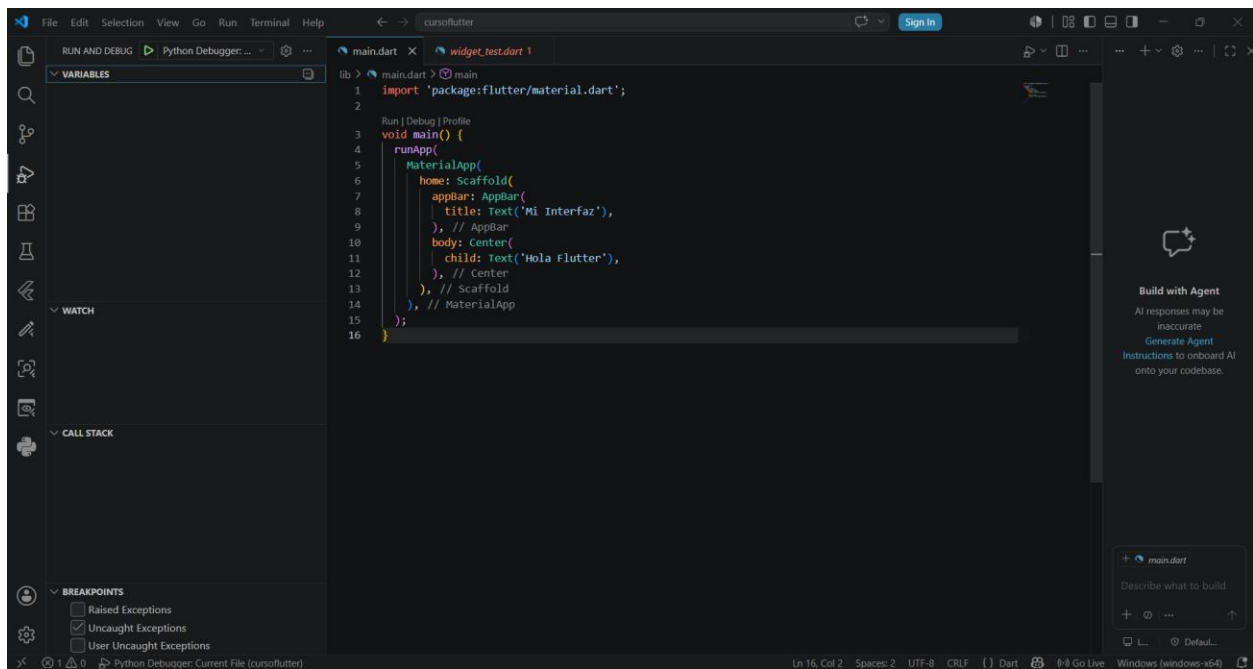
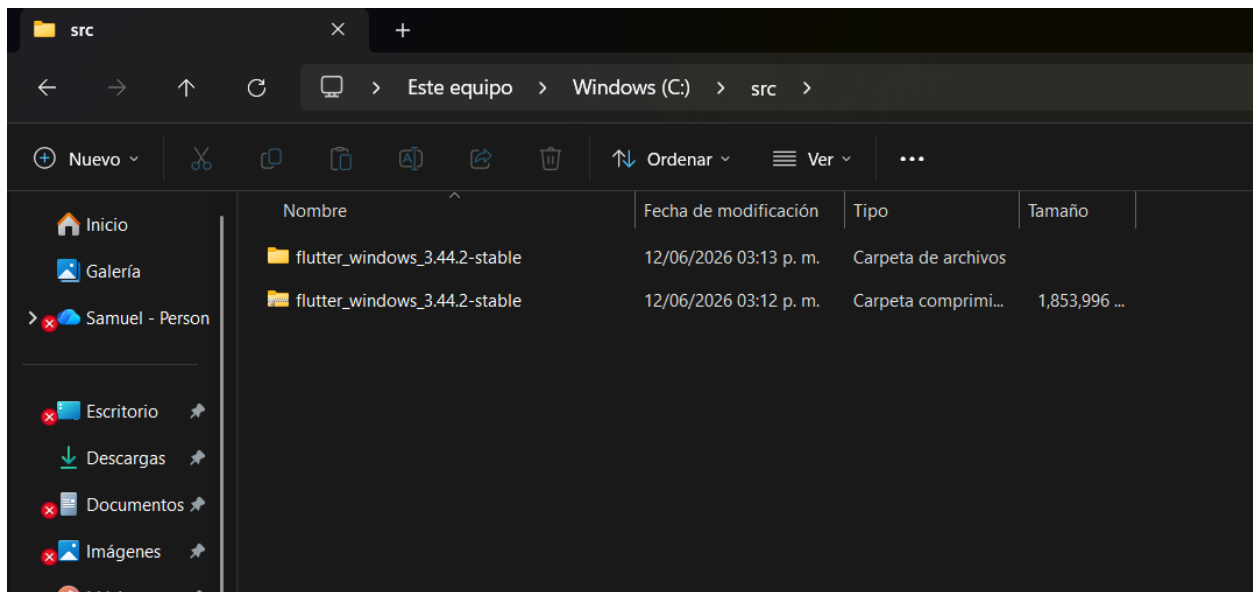
Seleccione una ubicación que no contenga caracteres especiales ni espacios en su ruta y que no requiera privilegios elevados.
- 3 Extraer el SDK**

Extrae el paquete SDK que descargaste en el directorio donde deseas almacenar el SDK de Flutter.

1. Copia el siguiente comando.

En esta página

- Elige tu plataforma de desarrollo
- Descargue el software necesario.
- Instalar y configurar Flutter
- Agrega Flutter a tu PATH
- Continúa tu viaje con Flutter



Tema 3. Primeros pasos (ejemplo 1)

```
main.dart x widget_test.dart 1
lib > main.dart > main
1 import 'package:flutter/material.dart';
2
Run | Debug | Profile
3 void main() {
4   runApp(
5     MaterialApp(
6       home: Scaffold(
7         appBar: AppBar(
8           title: Text('Mi Interfaz'),
9         ), // AppBar
10        body: Center(
11          child: Text('Hola Flutter'),
12        ), // Center
13      ), // Scaffold
14    ), // MaterialApp
15  );
16 }
```

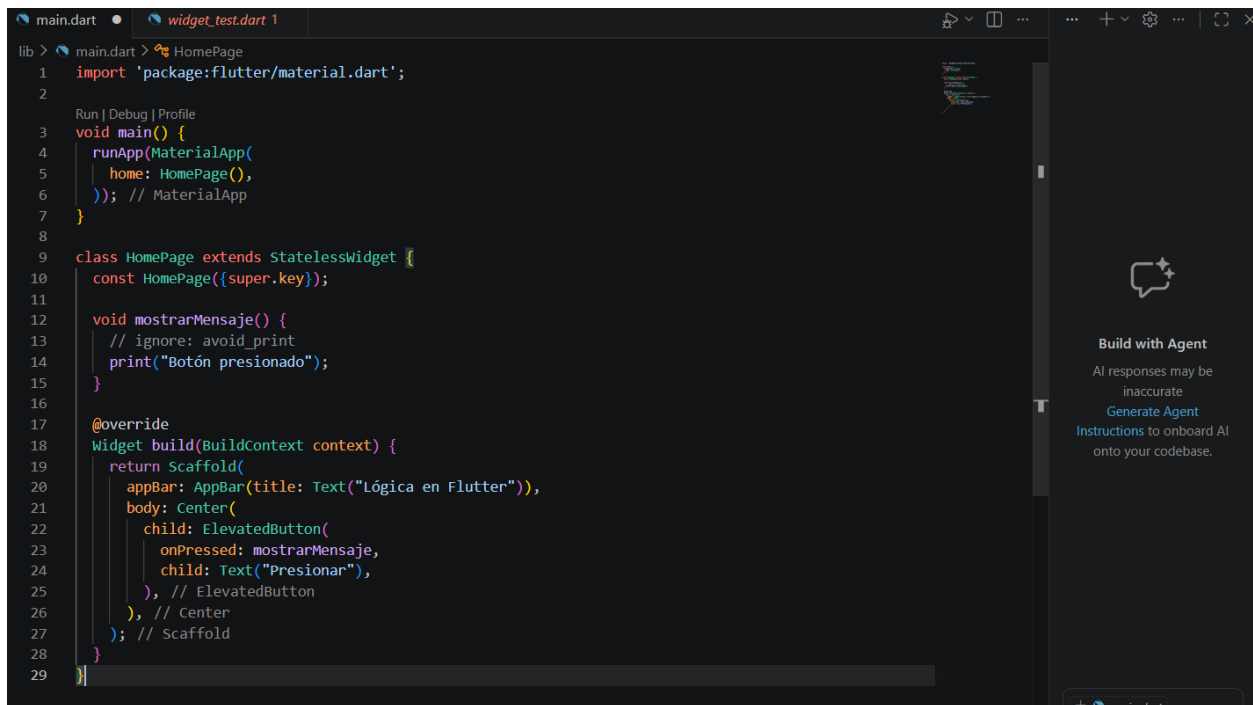
```
main.dart x widget_test.dart 1 x
test > widget_test.dart > ...
1 // This is a basic Flutter widget test.
2 //
3 // To perform an interaction with a widget in your test, use the WidgetTester
4 // utility in the flutter_test package. For example, you can send tap and scroll
5 // gestures. You can also use WidgetTester to find child widgets in the widget
6 // tree, read text, and verify that the values of widget properties are correct.
7
8 import 'package:flutter/material.dart';
9 import 'package:flutter_test/flutter_test.dart';
10
11 import 'package:cursoflutter/main.dart';
12
Run | Debug
13 void main() {
Run | Debug
14 testWidgets('Counter increments smoke test', (WidgetTester tester) async {
15   // Build our app and trigger a frame.
16   await tester.pumpWidget(const MyApp());
17
18   // Verify that our counter starts at 0.
19   expect(find.text('0'), findsOneWidget);
20   expect(find.text('1'), findsNothing);
21
22   // Tap the '+' icon and trigger a frame.
23   await tester.tap(find.byIcon(Icons.add));
24   await tester.pump();
25
26   // Verify that our counter has incremented.
27   expect(find.text('0'), findsNothing);
28   expect(find.text('1'), findsOneWidget);
29 });
30 }
31
```

3.1

```
main.dart x widget_test.dart 1
lib > main.dart > HomePage
1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(MyApp());
5 }
6
7 class MyApp extends StatelessWidget {
8   @override
9   Widget build(BuildContext context) {
10     return MaterialApp(
11       home: HomePage(),
12     ); // MaterialApp
13   }
14 }
15
16 class HomePage extends StatelessWidget {
17   @override
18   Widget build(BuildContext context) {
19     return Scaffold(
20       appBar: AppBar(
21         title: Text('Mi Aplicación'),
22       ), // AppBar
23       body: Center(
24         child: Text('Hola Flutter'),
25       ), // Center
26     ); // Scaffold
27   }
28 }
```

D/EGL_emulation(12805): eglCreateContext: 0xea305480: maj 3 min 1 rcv 4
D/EGL_emulation(12805): eglMakeCurrent: 0xea305480: ver 3 1 (tinfo 0xd3a7b3f0)
E/eglCodecCommon(12805): glUtilsParamSize: unknow param 0x000082da
E/eglCodecCommon(12805): glUtilsParamSize: unknow param 0x000082da
I/Choreographer(12805): Skipped 32 frames! The application may be doing too much work on its main thread.
D/EGL_emulation(12805): eglMakeCurrent: 0xea0051c0: ver 3 1 (tinfo 0xd3a7b070)
D/EGL_emulation(12805): eglMakeCurrent: 0xea305480: ver 3 1 (tinfo 0xd3a7b3f0)
D/skia (12805): Program linking failed.
I/chatty (12805): uid=10085(com.example.first_project) 1.gpu identical 7 lines
D/skia (12805): Program linking failed.

3.2



The screenshot shows an IDE with two tabs: `main.dart` and `widget_test.dart 1`. The `main.dart` file contains the following code:

```
lib > main.dart > HomePage
1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(MaterialApp(
5     home: HomePage(),
6   )); // MaterialApp
7 }
8
9 class HomePage extends StatelessWidget {
10   const HomePage({super.key});
11
12   void mostrarMensaje() {
13     // ignore: avoid_print
14     print("Botón presionado");
15   }
16
17   @override
18   Widget build(BuildContext context) {
19     return Scaffold(
20       appBar: AppBar(title: Text("Lógica en Flutter")),
21       body: Center(
22         child: ElevatedButton(
23           onPressed: mostrarMensaje,
24           child: Text("Presionar"),
25         ), // ElevatedButton
26       ), // Center
27     ); // Scaffold
28   }
29 }
```

On the right side of the IDE, there is a sidebar with the text "Build with Agent" and "AI responses may be inaccurate". Below this, there is a button labeled "Generate Agent" and a link "Instructions to onboard AI onto your codebase."

Tema 4 Widgets en común (ejemplos 2)



The screenshot shows an IDE with two tabs: `main.dart` and `widget_test.dart 1`. The `main.dart` file contains the following code:

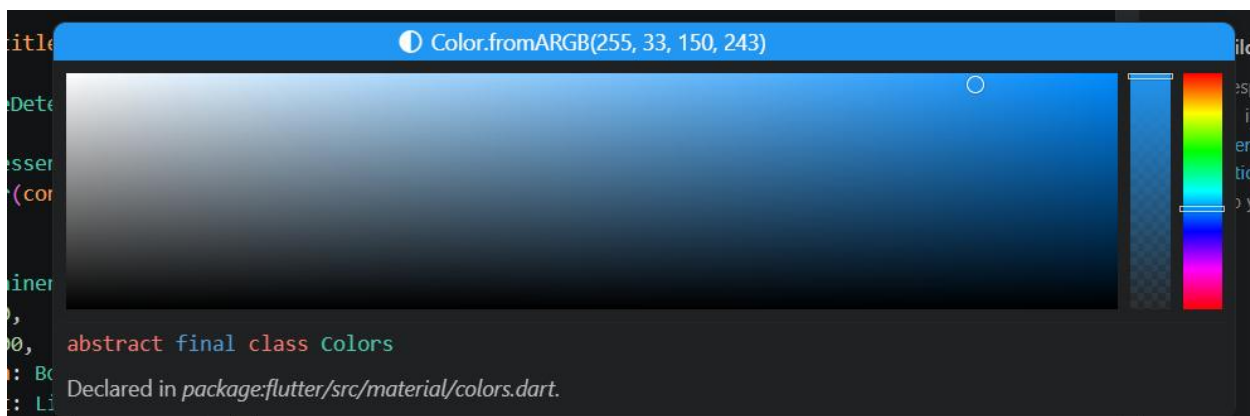
```
lib > main.dart > main
1 import 'package:flutter/material.dart';
2
3 void main() {
4   runApp(
5     MaterialApp(
6       home: Scaffold(
7         appBar: AppBar(
8           title: Text('Mi App'),
9         ), // AppBar
10        body: Center(
11          child: Icon(
12            Icons.favorite,
13            size: 80,
14          ), // Icon
15        ), // Center
16      ), // Scaffold
17    ), // MaterialApp
18  );
19 }
```

4.1

```
lib > main.dart > HomePage
1 import 'package:flutter/material.dart';
2
Run | Debug | Profile
3 void main() {
4   runApp(MaterialApp(
5     home: HomePage(),
6   )); // MaterialApp
7 }
8
9 class HomePage extends StatelessWidget {
10   @override
11   Widget build(BuildContext context) {
12     return Scaffold(
13       appBar: AppBar(title: Text('Flutter')),
14       body: Center(
15         child: GestureDetector(
16           onTap: () {
17             ScaffoldMessenger.of(context).showSnackBar(
18               SnackBar(content: Text('¡Container presionado!')),
19             );
20           },
21         ),
22       ),
23     );
24   }
25 }
```

```
21   child: Container(
22     width: 200,
23     height: 100,
24     decoration: BoxDecoration(
25       gradient: LinearGradient(
26         colors: [Colors.blue, Colors.purple],
27       ), // LinearGradient
28     ), // BoxDecoration
29     child: Center(
30       child: Text(
31         'Tócame',
32         style: TextStyle(color: Colors.white),
33       ), // Text
34     ), // Center
35   ), // Container
36 ), // GestureDetector
37 ), // Center
38 ); // Scaffold
39 }
40 }
```

Cambio de color



4.3

```
lib > main.dart > main
1 import 'package:flutter/material.dart';
2
Run | Debug | Profile
3 void main() {
4   runApp(MaterialApp(
5     home: Scaffold(
6       appBar: AppBar(
7         title: Text('Ejemplo Flutter'),
8       ), // AppBar
9       body: SingleChildScrollView(
10        child: Column(
11          children: [
12            Text(
13              'Bienvenido a Flutter',
14              style: TextStyle(fontSize: 24),
15            ), // Text
16            Row(
17              mainAxisAlignment: MainAxisAlignment.center,
18              children: [
19                Text('Uno'),
20                SizedBox(width: 20),
21                Text('Dos'),
22                SizedBox(width: 20),
23                Text('Tres'),
24              ],
25            ), // Row
26          ],
27        ), // Column
28      ), // SingleChildScrollView
29    ), // Scaffold
30  )); // MaterialApp
31 }
```

Tema 6. Trabajar con formularios

```
1 import 'package:flutter/material.dart';
2
Run | Debug | Profile
3 void main() {
4   runApp(MaterialApp(
5     home: FormExample(),
6   )); // MaterialApp
7 }
8
9 class FormExample extends StatelessWidget {
10   final TextEditingController nombre = TextEditingController();
11
12   @override
13   Widget build(BuildContext context) {
14     return Scaffold(
15       appBar: AppBar(
16         title: Text('Formulario'),
17       ), // AppBar
18       body: Padding(
19         padding: EdgeInsets.all(20),
20         child: Column(
```

```

21     children: [
22       TextField(
23         controller: nombre,
24         decoration: InputDecoration(
25           labelText: 'Nombre',
26         ), // InputDecoration
27       ), // TextField
28       SizedBox(height: 20),
29       ElevatedButton(
30         onPressed: () {
31           print(nombre.text);
32         },
33         child: Text('Enviar'),
34       ), // ElevatedButton
35     ],
36   ), // Column
37 ), // Padding
38 ); // Scaffold
39 }
40 }

```

```

1  import 'package:flutter/material.dart';
2  main()
Run | Debug > FormExample
3  void main() {
4    runApp(MaterialApp(
5      home: FormExample(),
6    )); // MaterialApp
7  }
8  ⚡

```

Tema 7. Navegación a través de las pantallas

```

lib > main.dart > Pantalla2
1  import 'package:flutter/material.dart';
2
Run | Debug | Profile
3  void main() {
4    runApp(MaterialApp(
5      home: Pantalla1(),
6    )); // MaterialApp
7  }
8
9  class Pantalla1 extends StatelessWidget {
10   @override
11   Widget build(BuildContext context) {
12     return Scaffold(
13       appBar: AppBar(title: Text('Pantalla 1')),
14       body: Center(
15         child: ElevatedButton(
16           child: Text('Ir a Pantalla 2'),
17           onPressed: () {
18             Navigator.push(
19               context,
20               MaterialPageRoute(

```

```

21         builder: (context) => Pantalla2(),
22       ), // MaterialPageRoute
23     );
24   },
25   ), // ElevatedButton
26   ), // Center
27 ); // Scaffold
28 }
29 }
30
31 class Pantalla2 extends StatelessWidget {
32   @override
33   Widget build(BuildContext context) {
34     return Scaffold(
35       appBar: AppBar(title: Text('Pantalla 2')),
36       body: Center(
37         child: Text('Bienvenido'),
38       ), // Center
39     ); // Scaffold
40   }
41 }

```

7.1

```

lib > main.dart > Pantalla2
1 import 'package:flutter/material.dart';
2
3 Run | Debug | Profile
4 void main() {
5   runApp(MaterialApp(
6     home: Pantalla1(),
7   )); // MaterialApp
8 }
9
10 class Pantalla1 extends StatelessWidget {
11   @override
12   Widget build(BuildContext context) {
13     return Scaffold(
14       appBar: AppBar(title: Text('Pantalla 1')),
15       body: Center(
16         child: ElevatedButton(
17           child: Text('Ir a Pantalla 2'),
18           onPressed: () {
19             Navigator.push(
20               context,
21               MaterialPageRoute(

```

```

21         builder: (context) => Pantalla2(),
22       ), // MaterialPageRoute
23     );
24   },
25   ), // ElevatedButton
26   ), // Center
27 ); // Scaffold
28 }
29 }
30
31 class Pantalla2 extends StatelessWidget {
32   @override
33   Widget build(BuildContext context) {
34     return Scaffold(
35       appBar: AppBar(title: Text('Pantalla 2')),
36       body: Center(
37         child: ElevatedButton(
38           child: Text('Regresar'),
39           onPressed: () {
40             Navigator.pop(context);
41           },
42         ), // ElevatedButton
43       ), // Center
44     ); // Scaffold
45   }
46 }

```

7.2

Se usa para reemplazar la pantalla actual por otra, sin dejar la pantalla anterior en la pila.

```

lib > main.dart > pushReplacementNamed
1 Navigator.pushReplacementNamed(
2   context,
3   '/segunda',
4 );

```

```

lib > main.dart > ElevatedButton
1 ElevatedButton(
2   onPressed: () {
3     Navigator.pushReplacementNamed(
4       context,
5       '/home',
6     );
7   },
8   child: Text('Ir a Home'),
9 );

```

Tema 8. Analisis de JSON

```
main.dart 1 • widget_test.dart 1
lib > main.dart
1 {
2   "nombre": "SamUEL",
3   "edad": 22
4 }
```

Import

```
main.dart • widget_test.dart •
lib > main.dart > main
1 import 'dart:convert';
2
Run | Debug | Profile
3 void main() {
4   String jsonData = '{"nombre":"SamUEL","edad":2}';
5
6   Map<String, dynamic> datos = jsonDecode(jsonData);
7
8   print(datos['nombre']);
9   print(datos['edad']);
10 }
```

8.1

```
lib > main.dart
1 {
2   "nombres": ["Samuel", "Juan", "Ana"]
3 }
```

```
lib > main.dart > ...
1 import 'dart:convert';
2
Run | Debug | Profile
3 void main() {
4   String jsonData =
5     '{"nombres":["Samir","Juan","Ana"]}';
6
7   Map<String, dynamic> datos = jsonDecode(jsonData);
8
9   List nombres = datos['nombres'];
10
11   print(nombres[0]);
12   print(nombres[1]);
13   print(nombres[2]);
14 }
15
```